

AI-Driven Protein Design with Diffusion and Language Models: A Survey

AI-driven protein design uses learned sequence, structure, and function models to propose proteins with desired folds, binding properties, catalytic activities, stability, or interaction profiles. Diffusion models have become powerful generators of backbones, motifs, binders, and complexes, while protein language models provide representations and generative priors over amino-acid sequences. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run and organizes the literature around protein language models, diffusion and score-based generation, structure prediction and geometric foundation models, inverse folding, function-aware conditional design, protein complexes and docking, optimization with experimental feedback, and validation. The survey argues that reliable protein design requires coupling generative priors with physical constraints, functional assays, uncertainty estimates, and responsible evaluation.

Keywords:

protein design; diffusion models; protein language models; inverse folding; structure generation; binder design; experimental validation; AI for biology

1 Introduction

Protein design asks for amino-acid sequences and three-dimensional structures that realize desired biochemical behavior. The design target may be a fold, a binding interface, an enzyme active site, a therapeutic antibody, a protein complex, or a property profile such as stability and solubility. AI-driven methods have changed the field by learning priors from large sequence and structure databases, then using those priors to generate, score, refine, and experimentally validate candidate proteins [90, 112, 132, 166, 230, 234, 242, 247].

Two modeling families are especially important. Protein language models learn statistical regularities over amino-acid sequences and can provide embeddings, masked-token predictions, generative sequence priors, and structure-aware representations. Diffusion and score-based models learn to denoise coordinates, frames, distances, torsions, or sequence states, making them natural tools for backbone generation, motif scaffolding, binder design, and conditional structure sampling [80, 112, 164, 199, 230, 234, 242, 247]. Their combination suggests a new design

loop: generate a structure or sequence, enforce constraints, score function and developability, redesign incompatible residues, and send a small set of candidates to experiments.

This survey reviews AI-driven protein design with diffusion and language models. We emphasize representation choices, generation objectives, geometric constraints, function conditioning, interaction modeling, optimization loops, and validation standards.

2 Background

2.1 Protein Sequence, Structure, and Function

Proteins are polymers whose sequence determines a folded structure and whose structure helps determine function. Design therefore couples a discrete sequence problem with a continuous geometry problem. A generated protein must be physically plausible, foldable, stable, soluble, specific to intended partners, and compatible with downstream assays. Structure prediction models and geometric neural networks provide useful constraints,

but function is often harder to predict than folding [90, 112, 120, 166, 224, 229, 256, 279].

2.2 Protein Language Models

Protein language models treat amino-acid sequences as biological text. Masked, autoregressive, and encoder-decoder objectives learn conservation patterns, remote homology, residue context, and functional motifs from large databases. These models support property prediction, mutation effect estimation, sequence generation, and representation learning. For design, the key question is whether a model captures only sequence likelihood or also useful constraints on folding and function [80, 112, 164, 199, 230, 234, 242, 247].

2.3 Diffusion Models for Molecular Geometry

Diffusion models generate samples by reversing a noising process. In protein design, the noised object may be atom coordinates, residue frames, pairwise distances, torsion angles, sequences, or hybrid sequence-structure states. Equivariance and physical constraints matter because rotations, translations, bond geometry, and chain connectivity should be respected. Diffusion methods can generate diverse backbones and condition on motifs or target interfaces [74, 90, 112, 132, 166, 230, 234, 247].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the task topic with protein design, de novo protein design, protein generation, diffusion models, score-based modeling, RFDiffusion, Chroma, protein language models, ESM, ProGen, ProtGPT2, ProtTrans, inverse folding, ProteinMPNN, fixed-backbone sequence design, motif scaffolding, binder design, enzyme and antibody design, protein complexes, docking, geometric deep learning, AlphaFold, RoseTTAFold, reinforcement learning, active learning, Bayesian optimization, directed evolution, fitness landscapes, benchmarks, wet-lab validation, and experimental feedback. Records were retained when title or abstract evidence connected proteins with design, generation, inverse folding, structure/-function modeling, diffusion, language models, optimization, or experimental validation. Duplicates

were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize the literature along eight axes. The first is *design object*: sequence, backbone, side chain, motif, binder, enzyme, antibody, peptide, complex, or protein-ligand interface. The second is *representation*: tokens, multiple-sequence alignments, residue graphs, coordinates, residue frames, distance maps, torsions, or all-atom structures. The third is *model family*: protein language model, diffusion model, geometric neural network, inverse-folding model, structure predictor, docking model, reinforcement learner, or Bayesian optimizer. The fourth is *conditioning*: target fold, motif, binding partner, active site, sequence family, property, text prompt, or experimental feedback. The fifth is *objective*: likelihood, denoising score matching, masked prediction, structure recovery, binding affinity, stability, diversity, novelty, or assay score. The sixth is *constraint*: symmetry, chain connectivity, steric validity, residue chemistry, developability, immunogenicity, and biosafety. The seventh is *evaluation*: structural plausibility, sequence recovery, designability, function prediction, experimental success, diversity, and failure analysis. The eighth is *deployment*: wet-lab cost, uncertainty, screening throughput, responsible release, and domain-specific validation.

3.3 Protein Language Models and Sequence Priors

Language models provide priors over protein families and remote sequence relationships. In design pipelines they can initialize sequences, score mutations, encode user-specified context, or supply embeddings to downstream predictors. Autoregressive models can generate novel sequences, while masked models can fill residues around structural or functional constraints. Their limitations include weak explicit geometry, sensitivity to training data bias, and uncertain mapping from likelihood to biological function [80, 112, 164, 199, 230, 234, 242, 247].

3.4 Diffusion and Score-Based Protein Generation

Diffusion methods are well suited to generating protein geometry because they can sample diverse structures while conditioning on motifs, targets, or constraints. Backbone diffusion supports fold generation and scaffold design; joint sequence-structure diffusion can coordinate residue identity with geome-

try; conditional diffusion can bias generation toward binders, enzymes, or complexes. Key design choices include the noise schedule, coordinate frame, equivariant architecture, conditioning channel, and post-generation filtering [74, 90, 112, 132, 166, 230, 234, 247].

3.5 Structure Prediction and Geometric Foundation Models

Structure predictors and geometric models are now central to design. They score whether generated sequences fold as intended, provide structural embeddings, and sometimes act as differentiable or black-box oracles. Equivariant networks and graph models encode residue geometry and interactions. However, high predicted confidence does not guarantee function, specificity, manufacturability, or experimental success [90, 112, 120, 166, 224, 229, 256, 279].

3.6 Inverse Folding and Sequence Design

Inverse folding maps a target structure or backbone to compatible sequences. It is important because many diffusion systems generate backbones first, then need sequences that stabilize those structures. Models such as fixed-backbone sequence designers can propose diverse candidates, but they must handle side-chain packing, surface properties, and functional residues. Sequence design also interacts with language-model priors and experimental selection [74, 112, 132, 164, 230, 234, 247, 256].

3.7 Function-Aware Conditional Design

Useful proteins are not merely foldable. They bind, catalyze, switch, assemble, or evade degradation under specific conditions. Function-aware design conditions generation on active sites, motifs, epitopes, ligands, target surfaces, or property predictors. The challenge is that functional labels are sparse and noisy compared with sequence databases, so models often combine learned priors with physics-inspired scoring, docking, and experimental screens [90, 112, 132, 164, 230, 234, 242, 247].

3.8 Complexes, Interactions, and Docking

Many design tasks are interaction tasks: binders, antibodies, peptides, protein complexes, and protein-ligand systems. Models must reason about interfaces,

shape complementarity, affinity, specificity, conformational change, and off-target binding. Diffusion and geometric models can generate complexes or interface-compatible backbones, while docking and structure prediction provide filters and refinement signals [90, 112, 120, 151, 164, 166, 229, 230].

3.9 Optimization and Experimental Feedback

Design becomes reliable when generation is connected to selection. Active learning, Bayesian optimization, reinforcement learning, directed evolution, and surrogate fitness models can prioritize candidates for assays. Experimental feedback then updates models or filters future designs. This loop is expensive, so uncertainty, diversity, and calibration matter as much as raw predicted score [74, 80, 112, 164, 199, 229, 230, 234].

3.10 Benchmarks, Validation, and Responsible Evaluation

Protein design evaluation spans computational and experimental evidence. Computational metrics include sequence recovery, structure prediction confidence, diversity, novelty, stability predictions, and docking scores. Experimental validation tests folding, expression, binding, activity, specificity, and developability. Benchmarks should avoid leakage from homologous sequences or structures and should report failure cases, wet-lab hit rates, and safety constraints [132, 151, 164, 166, 230, 234, 242, 256].

3.11 Representative Literature Map

The following map cites the retained corpus by theme, making the coverage explicit across protein language models, diffusion generation, geometric structure models, inverse folding, conditional function design, complexes, optimization, and validation.

Protein language models and sequence representations The protein language models and sequence representations cluster contains work that develops methods, systems, datasets, or analyses central to this survey [77, 80, 164, 199, 230, 234, 239, 242, 247]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [48, 93, 106, 128, 145, 152, 165, 277, 298]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [19, 21, 105, 117, 136, 179, 180, 207, 272]. Additional retained studies broaden the cluster with alter-

native architectures, training signals, data sources, and deployment assumptions [9, 11, 28, 45, 49, 86, 250, 257, 295]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [41, 113, 126, 138, 186, 213, 214, 266, 281]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [5, 26, 69, 114, 154, 185, 203, 209, 215]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [62, 89, 129, 137, 143, 147, 158, 195, 291]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [10, 25, 30, 52, 92, 169, 178, 196, 236]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [58, 144, 171, 181, 212, 231, 296].

Diffusion and score-based protein generation The diffusion and score-based protein generation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [38, 60, 74, 148, 149, 166, 217, 267, 294]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [81, 102, 107, 123, 133, 134, 139, 160, 237]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [13, 36, 79, 99, 121, 146, 184, 187, 261]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [20, 24, 50, 66, 135, 157, 167, 253, 273]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [37, 51, 76, 83, 190, 192, 241, 246, 259]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [98, 161, 175, 188, 216, 232, 268, 274, 276].

Structure prediction and geometric foundation models The structure prediction and geometric foundation models cluster contains work that develops methods, systems, datasets, or analyses central to this survey [110, 112, 124, 229, 251, 256, 262, 263, 279]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [53, 67, 68, 78, 153, 155, 211, 221, 260]. Additional retained studies broaden the cluster with alternative architectures, training

signals, data sources, and deployment assumptions [1, 2, 16, 23, 61, 142, 280, 282, 283]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [32, 59, 97, 163, 202, 204, 219, 278, 292]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [22, 33, 170].

Inverse folding and sequence design for target structures The inverse folding and sequence design for target structures cluster contains work that develops methods, systems, datasets, or analyses central to this survey [29, 42, 71, 85, 101, 109, 132, 159, 255]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [44, 73].

Function-aware and conditional protein design The function-aware and conditional protein design cluster contains work that develops methods, systems, datasets, or analyses central to this survey [3, 54, 120, 151, 174, 205, 228, 244, 286]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [40, 46, 88, 193, 208, 233, 243, 269, 285]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [14, 31, 35, 84, 108, 122, 194, 265, 289]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [47, 64, 75, 96, 141, 156, 191, 284, 290]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [57, 70, 125, 131, 197, 198, 218, 223, 270]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [63, 72, 94, 173, 183, 200, 275, 297].

Protein complexes, interactions, and docking The protein complexes, interactions, and docking cluster contains work that develops methods, systems, datasets, or analyses central to this survey [95, 115, 189].

Optimization, reinforcement learning, and experimental feedback The optimization, reinforcement learning, and experimental feedback cluster contains work that develops methods, systems, datasets, or analyses central to this survey [15, 27, 100, 103, 116, 140, 226, 235, 252]. A second strand in this

cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [104, 172, 210, 271].

Benchmarks, validation, safety, and applications

The benchmarks, validation, safety, and applications cluster contains work that develops methods, systems, datasets, or analyses central to this survey [7, 39, 43, 65, 90, 176, 224, 254, 300]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [17, 87, 127, 150, 206, 220, 225, 238, 240]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [55, 162, 168, 177, 182, 201, 227, 258, 264]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [4, 18, 82, 118, 222, 248, 249, 287, 293]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [8, 56, 91, 111, 119, 130, 245, 288, 299]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [6, 12, 34].

4 Challenges and Prospects

4.1 From Plausible Structures to Verified Function

Generative models can produce plausible backbones more easily than verified function. Future systems need stronger links among structure, dynamics, binding, catalysis, expression, and assay outcomes.

4.2 Hybrid Sequence-Structure Generation

Protein design is neither purely textual nor purely geometric. Promising systems will coordinate sequence tokens, residue frames, side chains, constraints, and functional annotations in one generation-and-refinement loop.

4.3 Uncertainty and Experimental Allocation

Wet-lab validation is costly. Models should report uncertainty, novelty, and diversity so that experiments test informative candidates rather than only the highest-scoring outputs of a possibly biased oracle.

4.4 Data Leakage and Benchmark Realism

Protein databases contain homologs, designed variants, and benchmark overlaps. Evaluation should use temporal and homology-aware splits and distinguish rediscovery from genuinely novel design.

4.5 Biosafety and Responsible Release

Powerful protein generators can support therapeutics, enzymes, and materials, but they also raise safety concerns. Responsible development requires screening, access control where appropriate, transparent limitations, and evaluation of dual-use risks.

5 Conclusion

AI-driven protein design with diffusion and language models is becoming a practical design paradigm. Protein language models supply sequence priors, diffusion models generate structures and condition on motifs or interfaces, geometric models connect sequence to fold, and optimization loops connect computation to experiments. The central lesson is that successful protein design requires integration: generative models, physical constraints, function predictors, uncertainty, and wet-lab validation must operate together rather than as isolated modules.

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